PROJECT REPORT

PhoneSploit-Pro

Android Penetration Testing & ADB Automation Tool

Submitted By

Name: Sk Ahnaf Labib

Institution: Southeast University

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1. Introduction

PhoneSploit-Pro is a Python-based Android penetration testing and ADB automation tool designed for cybersecurity researchers, ethical hackers, and learners. The tool uses Android

Debug Bridge (ADB) to communicate with Android devices and perform various automation and testing operations.

The project aims to simplify Android security testing and provide an all-in-one toolkit for device management and penetration testing activities.

PhoneSploit-Pro is an open-source project hosted on GitHub and developed mainly for educational and authorized security testing purposes.

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2. Objectives

The main objectives of this project are:

- To automate Android penetration testing tasks
- To simplify ADB device management
- To provide remote Android device control
- To support cybersecurity learning and research
- To create an easy-to-use ethical hacking toolkit

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3. Technologies Used

Technology	Purpose
Python	Main programming language
ADB (Android Debug Bridge)	Android device communication
Metasploit Framework	Penetration testing integration
Nmap	Network scanning
Scrcpy	Android screen control
Git & GitHub	Version control and project hosting

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4. Features of the Tool

Core Features

- Remote ADB Connection
- Android Device Detection
- Device Management
- Remote Shell Access
- Screenshot Capture
- Screen Recording
- APK Installation
- File Transfer Support
- App Management
- Device Reboot Options

Advanced Features

- Metasploit Integration
- Payload Generation
- Meterpreter Session Support
- Network Scanning
- Wireless Android Device Control

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5. System Requirements

The following software and tools are required to run PhoneSploit-Pro:

- Python 3.10 or higher
- pip package manager
- Android ADB Platform Tools
- Metasploit Framework
- Nmap
- Scrcpy

Supported Operating Systems

- Linux
- Windows
- macOS
- Termux (Android)

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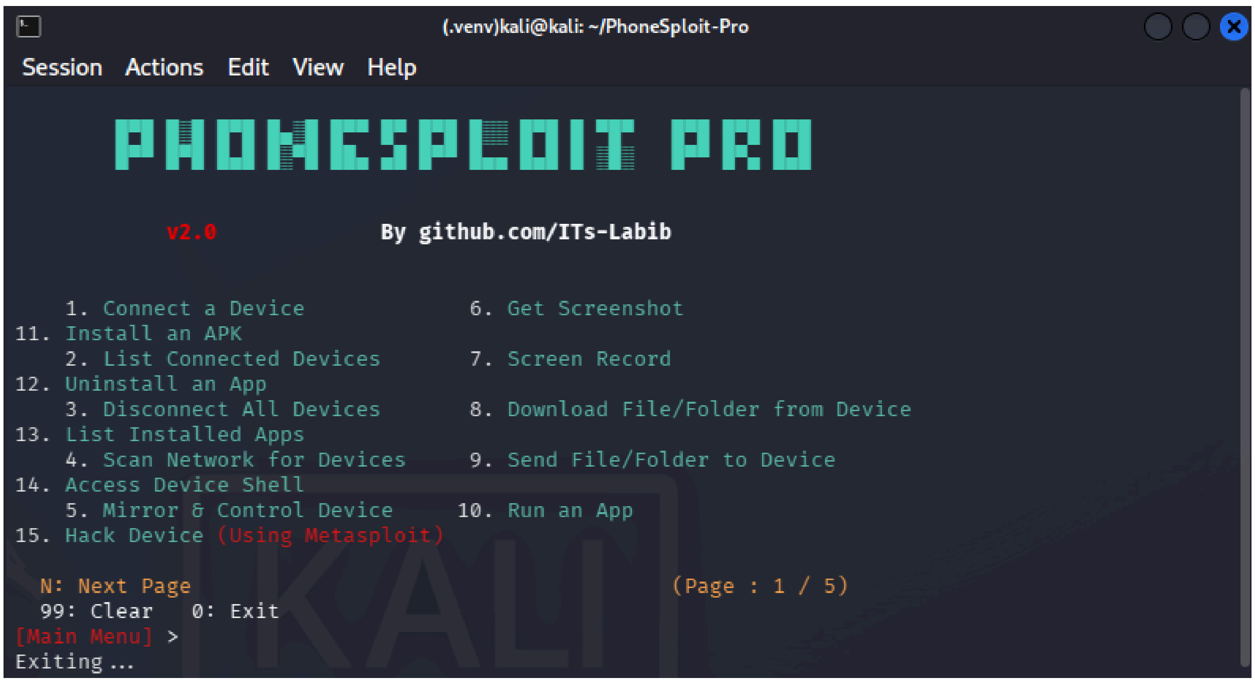
6. Working Procedure

The tool works through the following steps:

1. The user connects the Android device using ADB.
2. The tool detects the connected device.
3. Available operations and modules are displayed.
4. The user selects required testing or automation options.
5. The tool performs the selected operation automatically.
6. Advanced testing can be performed using Metasploit integration.



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7. Advantages

- Easy-to-use command-line interface
- Cross-platform compatibility
- Automation support
- Open-source availability
- Useful for cybersecurity education

- Supports multiple Android management operations

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8. Limitations

- USB Debugging must be enabled on the target device
- Some features work best on Linux systems
- Requires external dependencies such as Metasploit
- Unauthorized usage is illegal

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9. Ethical & Security Considerations

PhoneSploit-Pro should only be used for:

- Educational purposes
- Authorized penetration testing
- Security research
- Ethical hacking practice

Unauthorized access to devices or networks without permission is illegal and may violate cybersecurity laws.

Users must follow ethical hacking principles and ensure proper authorization before performing any testing activities.

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10. Future Improvements

Possible future improvements for the project include:

- Graphical User Interface (GUI)
- Cloud Integration
- Automated Report Generation
- Improved Android Version Compatibility
- AI-based Vulnerability Detection
- Enhanced Security Auditing Features

11. Conclusion

PhoneSploit-Pro is a powerful Android penetration testing and ADB automation framework that simplifies Android device management and security testing.

The project provides valuable learning opportunities for cybersecurity students, ethical hackers, and researchers while promoting automation in penetration testing activities.

As an open-source project, it can continue to evolve with additional features and community contributions.

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12. References

1. GitHub Repository:
   https://github.com/ITs-Labib/PhonesSploit-Pro.git
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3. Python Documentation
   https://docs.python.org/
4. Android Debug Bridge Documentation
   https://developer.android.com/tools/adb
5. Metasploit Framework Documentation
   https://docs.metasploit.com/
6. Nmap Documentation
   https://nmap.org/docs.html